Code Conversion

- There are various ways to represent the data.

  Characters are normally stored in ASCII format.

  Numbers can be represented by : binary, decimal, octal or hexadecimal systems.

$(10)_{hex} \Rightarrow 10/0A \Rightarrow$ quotient

Convert Hex(Binary) to BCD

- If the number is an 8 bit binary number then divide the number by 100d.
- The quotient forms the most significant BCD number.
- The remainder which we get should be divided by 10d.
- The quotient forms the next BCD number and the remainder forms the least significant BCD number.

Convert BCD to hexadecimal

(16)d ⇒ 1*0AH= (0A)

0A+6⇒ 10 it will be stored like this.

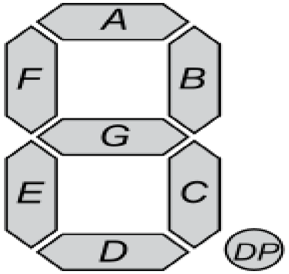
Convert BCD to hexadecimal

- First separate the BCD digits.
- The most significant digit will be multiplied with 0AH (10D)
- The least significant digit will be added with the result of multiplication.

Convert BCD to ASCII

- Separate the packed BCD numbers to unpacked BCD numbers.
- Add 30H to each unpacked BCD number.

Unpacked BCD number to 7 segment



- Numbers can be coded as:

Num	dp	g	f	e	d	c	b	a	Hex Equi
0	0	0	1	1	1	1	1	1	3F
1	0	0	0	0	0	1	1	0	06
2	0	1	0	1	1	0	1	1	5B
3	0	1	0	0	1	1	1	1	4F

- Similarly the rest of the numbers can be coded as:
  4-> 66H, 5-> 6DH, 6-> 7DH, 7-> 07H, 8-> 7FH,9-> 6FH

Program control Instructions

Mnemonics	Explanation
STC	Set CF $\leftarrow$ 1
CLC	Clear CF $\leftarrow$ 0
CMC	Complement carry CF $\leftarrow$ CF$^{/}$
STD	Set direction flag DF $\leftarrow$ 1
CLD	Clear direction flag DF $\leftarrow$ 0
STI	Set interrupt enable flag IF $\leftarrow$ 1
CLI	Clear interrupt enable flag IF $\leftarrow$ 0
NOP	No operation
HLT	Stops the execution of software.To exit from halt can be done either from interrupt/hardware reset.
WAIT	Wait for TEST pin active
ESC opcode mem/ reg	Used to pass instruction to a coprocessor which shares the address and data bus with the 8086
LOCK	Lock bus during next instruction

LAHF and SAHF instruction

- LAHF : Load AH with lower byte of flag register

  (AH)<- lower byte of flag register.

- SAHF : Store AH to to lower byte of flag register.

  lower byte of flag register <- (AH)

ORG and EVEN directives

- **ORG (Origin) : is used to assign the starting address (Effective address) for a program/ data segment**
- **Even: Informs the assembler to store program/ data segment starting from an even address**